RKDF UNIVERSITY, BHOPAL

B.Sc.(Electronics)

Scheme

Semester – I- VI

No	Subject Code	Subject Title	Marks Allotted						
			Assignment Marks		Theory Marks		Practical Marks		Total Marks
			Max	Min	Max	Min	Max	Min	
1	BEC 111	FUNDAMENTALS OF ELECTRONICS	**20**	**8**	**80**	**27**	**50**	**17**	**150**
2	BEC 121	SEMICONDUCTOR DEVICES AND IC FABRICATION TECHNOLOGY	**20**	**8**	**80**	**27**	**50**	**17**	**150**
3	BEC 131	ELECTRONIC COMMUNICATION SYSTEMS	**20**	**8**	**80**	**27**	**50**	**17**	**150**
4	BEC 141	MICROPROCESSOR AND ITS APPLICATIONS	**20**	**8**	**80**	**27**	**50**	**17**	**150**
5	BEC 151	COMMUNICATION	**20**	**8**	**80**	**27**	**50**	**17**	**150**
6	BEC 161	ELECTRONIC INSTRUMENTATION, DIGITAL SYSTEM DESIGN &MICROCONTROLLER	**20**	**8**	**80**	**27**	**50**	**17**	**150**

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B.Sc.(Electronics)
Syllabus
Semester – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Electronics)	**FUNDAMENTALS OF ELECTRONICS**	**BEC 111**

UNIT-I

Diode circuits and power Supplies: Junction diode characteristics - Half and full wave rectifiers - Expression for efficiency and ripple factor - Construction of low range power peak using diodes - Bridge rectifier - Filter circuits - Zener Diode -Characteristics - Regulated power supply using Zener diode - Clipper and Clamper using diodes.Differentiator and integrator using resistor and capacitor.

UNIT-II

Transistor circuits: Characteristics of a transistor in CB, CE modes - Relatively merits - Graphical analysis in CE configuration - Transistor as a amplifier - RC coupled B.Sc. Electronics : Syllabus (CBCS) Single stage amplifier - Frequency response - Thevenin's and Norton's theorems - h parameters.Basis logic gates AND, OR, and NOT - Construction of basic logic gates using diodes and transistors.

UNIT-III

Amplifiers: General principles of small signal amplifiers - Classifications - RC Coupled amplifiers - Gain - Frequency response - Input and output impedance - Multistage amplifiers - Transformer coupled amplifiers - Equivalent circuits at low, medium and high frequencies – Emitter follower.Class A and Class B power amplifiers - Single ended and push-pull configurations -Power dissipation and output power calculations.

UNIT-IV

Feedback Amplifiers: Basic concept of feedback amplifiers - Transfer gain with feedback - General characteristics of negative feedback amplifier - Effect of negative feedback on gain - Gain stability - Distortion and bandwidth - Input and output resistance in the case of various types of feedback - Analysis of voltage and current in feedback amplifier circuits.

UNIT-V

Operational Amplifiers: Principles - Transfer characteristics - Various offset parameters - Differential gain - CMRR - Slew rate - Bandwidth.Op-amp Circuits: Basic operational amplifier circuits under inverting and noninverting modes - Adder - Subtractor - Integrator - ifferentiator - Comparator -Sine, square and triangular waveform generators - Active filters - Sample and Hold .circuits.Oscillators: Positive feedback - Stability issues - Feedback requirement of oscillations -Barkhausen criterion for oscillation - Hartley, Colpitts, Phase shift and Wien bridge oscillators - Condition for oscillation and frequency derivation - Crystal oscillator -UJT relaxation oscillator.Monostable, bistable and astable multivibrators - Schmitt trigger.

Text Books

1. Electricity and Magnetism - M. Narayanamoorthi and Others, National Publishing Co., Chennai.

2. Electricity and Magnetism - R. Murugeshan, S. Chand & Co. Ltd., New Delhi,
3. Revised Edition, 2006.
4. Principles of Electronics - V.K. Mehta, S. Chand & Co., 4/e, 2001.
5. Basic Electronics - B.L. Theraja, S. Chand & Co., 4/e, 2001.
1. Fundamentals of Electricity and Magnetism - B.D. Duggal & C.L. Chhabra, Shoban Lal Nagin Chand & Co., Jallundur.
2. Physics, Vol. II - Resnick, Halliday & Krane, 5/e, John Wiley & Sons, Inc.,.
3. Basic Electronics - B. Grob, McGraw - hill, 6/e, NY, 1989.
4. Elements of Electronics - Bagde & Singh, S. Chand & Co. B.Sc. Electronics : Syllabus (CBCS)
1. Basic Electronics - A Text Lab Manual - Zbar, Malvino & Miller - Tata McGraw Hill.
2. B.E.S. Practicals - R. Sugaraj Samuel & Horsley Solomon - Department of Electronic
6. Science, C.T.M. College of Arts and Science, Chennai.
3. A Text Book of Practical Physics - M.N. Srinivasan & others - Sultan Chand & Sons, New Delhi.
4. Practical Physics - St.Joseph's College, Tiruchirappalli.
5. Practical Physics - M. Arul Thalapathi, Comtek Publishers, Kanchipuram
6. Linear Integrated Circuits - D. Roy Choudhury & Shail Jain, New Age International
7. (P) Limited.B.Sc. Electronics : Syllabus (CBCS)

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Syllabus

Semester – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Electronics)	SEMICONDUCTOR DEVICES AND IC FABRICATION TECHNOLOGY	BEC 121

UNIT-I

Digital Electronics

Analog and digital signals - Digital circuits - Binary number system - conversion of Binary to decimal - decimal to binary - logic gates - OR gate - AND gate - NOT gate - Combination of Logic gates - NAND and NOR as universal building blocks.

UNIT-II

Number system and codes: Decimal, binary, octal, hex numbers, conversion from one to another - codes, BCD, excess 3, gray codes conversion from one to another -Error correction / detection codes.Boolean algebra and theorems: Basic, Universal logic gates - Boolean Theorems - sum of products, products of sums expression, simplification by Karnaugh Map method, simplification based on basic Boolean theorems - don’t care conditions.Combinational Digital Circuits: Arithmetic building blocks, Basic Adders and Subtractors, BCD adders - Data processing circuits, multiplexers, demultiplexers, encoders, decoders - TTL, CMOS digital logic families.

UNIT-III

Sequential Digital Circuits: Flip - Flops, RS, clocked SR, JK, D, T, master-slave types -shift registers, ring counters-ripple counters - Design of counters - modulus of counters - timer IC 555, applications.DAC and ADC: Parameters, Accuracy, Resolution - DAC, variable resistor network, R-2R ladder network types - ADC, counting, continuous, successive approximation, dual-slope types - comparison of various types of ADC and DAC.

UNIT-IV

Transistors - Working of PNP and NPN transistors - Transistor connections -Relation between β and α - Expression for collector current - Transistor characteristics in CE mode - Transistor as an amplifier and oscillator its performance -Semiconductor devices numbering system - Phototransistor.Construction, working characteristics of FET and MOSFET (D and E type) -Parameters of FET - Difference between FET and BJT - Difference between FET and MOSFET - Applications of FET and MOSFET - Advantages of MOSFET.

UNIT-V

Construction, working characteristics of UJT and SCR I Equivalent circuit of UJT -SCR as a switch and rectifier - Applications of UJT and SCR - Characteristics of TRIAC. Schottky effect - Working characteristics of MIS, MIM diodes - Working and merits of CCD, LED and LCD - LDR - Photodiode - Solar cell - Semiconductor LASER diode and its application.Integrated circuit - Monolithic Integrated Circuit technology - Fabrication of IC components - Resistors, Capacitors, Diodes, Transistors, FET and MOSFET - Thin and thick film technology - LSI - MSI - VLSI - IC package and symbols - Merits and demerits of ICs.

Text Books

1. Electronic Devices and Circuits (Applied Electronics Vol. I) - G.K. Mithal, Khanna
2. Electronic Principles - A.P. Malvino, Tata McGraw Hill Publishing Co. Ltd.
3. A Text Book of Applied Electronics - R.S. Sedha, S. Chand & Co.,
4. Digital Fundamentals - V. Vijayendran, S.Viswanathan Publishers, Chennai.
5. Modern Digital Electronics - R.P. Jain, 2/e, Tata McGraw Hill Publishing Co. Ltd., New Delhi.
6. Micro Electronics - J. Millman, McGraw Hill International Book Company, New Delhi, 1990.
7. Digital Principles and Applications - A.P.Malvino & D.P.Leach, 4/e, Tata McGraw Hill Publishing Co. Ltd.
8. Digital Integrated Electronics - H. Taub & D. Schilling, McGraw-Hill Book Company.
9. Digital Fundamentals - T.L. Floyd, Pearson Education, 8/e.
10. Digital Electronics - W.H. Gothmann, Prentice Hall of India Private Limited, 2/e.B.Sc. Electronics : Syllabus (CBCS).

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Syllabus

Semester – III

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Electronics)	ELECTRONIC COMMUNICATION SYSTEMS	BEC 131

UNIT-I

Basic communication systems:

Block diagram - information source and input transducer - Transmitter medium -Noise - Receiver - Destination - Necessity for modulation - Types of communication systems. Amplitude Modulation: Definition - AM waveforms - Frequency spectrum and hand width - Modulation index - DSB - SC, SSB, Independent SB, Vestigial SB - Comparison and application of various AM schemes.

UNIT-II

Modulation - Needs for Modulation - Types of Modulation - Amplitude Modulation - Generation and detections circuits - Balanced Modulator - DSB/SC and SSB Modulation - VSB modulation. Block diagram of AM Radio transmitter and super heterodyne Receiver.Frequency Modulation - Definition - Derivation of Modulated wave - Generation of FM - Varactor diode and Reactance tube Modulators - Detectors - Balanced slope detector, Foster Seeley discriminator, ratio detector - Block diagram of FM transmitter and receiver.

UNIT-III

Pulse Modulation - Sampling theorem - PAM, PWM, PPM, PCM - quantizing, sampling, coding, decoding, quantization error, delta modulation and adaptive delta modulation. Multiplexing - FDM, TDM, CDMA - ASK, FSK, PSK - Advantages of Digital Communication - Introduction to Microwave, Fiber optic, Satellite Communications - RADAR - range equation.Antenna - Effective resistance - Efficiency - Directive gain - Bandwidth, Beam width and polarization - Dipole - Folded dipole - Arrays - Yagi - Uda - Helical - Discone -Parabolic - Dish Antennas - Ground wave, sky wave and space ware propagation - Skip distance - Maximum usable frequency.

UNIT-IV

Frequency and phase modulation: Definition - Relationship between FM & PM - Frequency deviation - Spectrum and transmission BW of FM, comparison of AM and FM systems.UNIT-IV: Radio Transmitter and Receiver AM transmitters - High level and low level transmitters - SSB transmitters - FM transmitters - Block diagram - stereo FM transmitter.AM receivers - operation - performance parameters - Communication Transceivers -Block diagram - SSB receiver - FM receivers - Block diagram.

UNIT-V

Television of TV system - Block diagram - Scanning - Synchronisation - VSB transmission and reception Colour signal transmission.

Text Books

1. Basics of electronic Communications NIIT, prentice - Hall Pvt. Ltd, New Delhi, 2007
2. Modern digital and analog communications - BP lathi third edition 1998, Oxford University press
3. Communication System: Analog & digital Singh and sapre, TMH 1995.
4. Electronic Communication Systems - George Kennedy, McGraw Hill Book Company, 4/e, 2005.
5. Communication Engineering - T.G. Palanivelu, Anuradha Publicatons, 1/e, 2002.
6. Communication System - Roddy & Coolen, 4/e, Pearson Education, 2005.
7. Principles of Communication Engineering - Anok Singh, 4/e, Sathyaprakasam Publications, 2004.
8. Electronic Communication Systems Wayne Tomasi, 4/e, Pearson Education, 2004.

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Syllabus
Semester – IV

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Electronics)	MICROPROCESSOR AND ITS APPLICATIONS	BEC 141

UNIT-I

Architecture of 8085 microprocessor - Registers - Flags - ALU - Address and data buses - Demultiplexing the address / data bus - Control and status signals -I0nstruction set of 8085 - Addressing modes - Assembly language programming -Programs for addition, subtraction, multiplication and division of binary and BCD numbers (8-bit only).

UNIT-II

Stack and stack related instructions - Subroutines - Advanced programming techniques - Code conversions - Block transfer of data - Sorting of data - Time delays using single register and register pair - Delay calculations.

UNIT-III

Semiconductor memories - Classification - Instruction cycle, Machine cycle and Tstate - Timing diagrams for opcode fetch, memory read, memory write, I/O read and I/O write machine cycles - Interfacing memory chips - Interfacing an input port -Interfacing an output port - I/O mapped I/O and memory mapped I/O techniques.B.Sc. Electronics : Syllabus (CBCS)

UNIT-IV

Interrupts - Hardware and software interrupts - Interrupt priorities - SIM and RIM instructions - Polled I/O and interrupt controlled I/O data transfer - Interfacing programmable devices - Programmable Peripheral Interface 8255 - Internal architecture - Control register and control word - Programming 8255 - Interfacing hex-keyboard and seven segment display.

UNIT-V

Interfacing D/A converter and waveform generation - Interfacing A/D converters -Keyboard / Display Controller 8279 - Internal architecture and working -Programmable Interval Timer 8253/54 - Internal architecture and different modes of operation - Stepper motor interface - Temperature controller - Traffic lights controller.

Text Books

1. Microprocessor Architecture, Programming and Applications with the 8085 -Ramesh S. Gaonkar, 5/e, Penram International Publishing (India).
2. Fundamentals of Microprocessors-8085 - V. Vijayendran, S. Viswanathan (Printers & Publishers), Pvt. Ltd., 2002.
3. Microprocessor and its Applications - A. Nagoor Kani, 1/e, RBA Publications, Chennai.
4. Introduction to Microprocessors - Aditya P. Mathur, 3/e, Tata McGraw Hill Publishing Company Limited.
5. Fundamentals of Microprocessors and Microcomputers - B. Ram, Fifth Revised and Enlarged Edition, Dhanpat Rai Publications, New Delhi.

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Syllabus
Semester –V

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Electronics)	COMMUNICATION	BEC 151

UNIT-I

Advanced mobile phone service - Global system for mobile communication - Digital cellular system - Cordless telephony - Third generation wireless systems. Cell structure - Hand off - roaming management - Hand off detection - Channel assignment techniques - Interference - ACI, CCI - Intersystem hand off and authentication - Network signaling - Cellular digital packet data. GSM - Network signaling, mobility management, short message service - International roaming, administration and operation.

UNIT-II

Wireless application protocol - Architecture - Datagram - Transport layer securities -Transaction protocol - Session protocol application environment, wireless markup language, WML - Script wireless telephony applications. Third generation mobile services - Wireless local loop - Bluetooth technology. Characteristics of Human eye - Theory of scanning - Camera tubes - Vidicon - Silicon diode array vidicon - Picture tubes - Composite video signal.

UNIT-III

Television transmitters - Television signal propagation - Television transmission antennas - Television receiver antennas - Colour Television Antennas - Television receiver - VHF Tuner - IF Subsystems - Video amplifiers - Sync processing and AFC circuit - Deflection oscillators. Colour Television systems - Colour characteristics - Colour Television Camera -Colour picture tube - Colour signal generation - PAL, NTSC, SECAM - Comparison.

UNIT-IV

Colour Television receivers - PAL D Colour receiver, AGC, Sync - Separators and deflection circuits, Luminance channel, Colour signal processing , separation of U and V modulation products - Subcarrier generation and control. Special Topics in Television - Digital tuning techniques - Remote control - Cable Television - Satellite TV - video tape recorders - Video disc systems - Digital TV -Fundamentals of Digital TV.

UNIT-V

Power Semiconductor Devices: Power diode, Power transistor, TRIAC, MOSFET and IGBT - turn on methods, driver circuits - SCR characteristics - Two transistor analogy - Methods of turning ON and turning OFF - Series and parallel connections of SCRs.Phase controlled converters: Single phase controlled rectifier - Half wave controlled rectifier with 1.Resistive load 2.RL load 3. RL load and battery - Full wave controlled rectifier with above types of loads - Three phase controlled rectifier - HVDC transmission. Inverters: Single phase and three phase inverters - Series and parallel inverters - Bridge inverters - Current source inverter. Choppers and Cycloconverters: Various types of DC choppers - Step up chopper -AD chopper - Single phase AC chopper - Step up and step down cycloconverters -Three phase to single phase and three phase to three phase cycloconverters.Control circuits and application: Generation of control pluses - Microprocessor based implementation - Static circuit breakers for DC and AC circuits - Regulated power

Text Books

1) Power Electronics - M.H. Rashid, Prentice Hill of India Private Limited.
2) Power Electronics - P.C. Sen, Tata McGraw Hill Publishing Co. Ltd.
3) Thyristorised Power Controllers - G.K. Debye, Wiley Eastern Ltd.
4) 2. An Introduction to Thyistors and Their Applications - M. Ramamoorthy, 2/e, East West press.
5) Mobile Communications - Jochen Schiller, 7/e, Pearson Education, 2003.
6) Principles of Wireless Networks - Kauch Pahalavan & Prahanet Krishnamoorthy, 2/e, Pearson Education, 2004.
7) Wireless and Mobile Networks Architecture - Yi-Bing Lin & Imnch Chlantee, John Wiley, 2001.
8) Wireless and Mobile Communication - Rapparport, Pearson Education, 2001.
9) Television and Video Engineering - G. Nagarajan, 2/e, A.R.S Publications, 2005.
10) Monochrome and Colour Television - R.R. Gulati, 1/e, New Age International Publishers, 2003.
    Basic Television - Principles and Servicing - Bernard Grob, 4/e, McGraw Hill, 1975.
    Television and Video Engineering - A. M. Dhake, 2/e, Tata McGraw Hill

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B.Sc.(Electronics)
Syllabus
Semester –VI

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Electronics)	ELECTRONIC INSTRUMENTATION, DIGITAL SYSTEM DESIGN &MICROCONTROLLER	BEC 161

UNIT-I

DC and AC indicating Instruments: Accuracy and precision - Types of errors -PMMC galvanometer, sensitivity, Loading effect - Conversion of Galvanometer into ammeter, Voltmeter and Shunt type ohmmeter- Multimeter. Electrodynamometer - Thermocouple instrument - Electrostatic voltmeter - Watt-hour meter.DC and AC bridges: Wheatstone bridge - Kelvin's bridge - Balancing condition for AC bridge - Maxwell's bridge - Schering's bridge - Wein's bridge - Determination of frequency.Oscilloscopes: Block diagram - Deflection Sensitivity - Electrostatic Deflection -Electrostatic Focusing - CRT Screen - Measurement of Waveform frequency, phase difference and Time intervals - Sampling Oscilloscope - Analog and Digital Storage Oscilloscopes.

UNIT-II

Instrumentation Amplifiers and Signal Analysers: Instrumentation amplifier -Electronic Voltmeter and Multimeter - Digital Voltmeter - Function Generator -Wave Analyser - Fundamentals of Spectrum Analyser.Transducer and Display Devices: Strain Gauge - Unbounded Strain Gauge - LVDT -Resistance Thermometer - Photoelectric Transducer - Pen Recorder - Audio Tape Recorder - Seven Segment Display - LCD.

UNIT-III

Boolean Algebra and Logic Gates: Review of binary number systems - Binary arithmetic - Binary codes - Boolean Algebra and theorems - Boolean functions -Simplifications of Boolean functions using Karnaugh map and tabulation methods -Logic gates.Combinational Logic: Combinational circuits - Analysis and design procedures -Circuits for arithmetic operations - Code conversions - Introduction to Hardware Description Language (HDL).Design with MSI Devices: Decoders and Encoders - Multiplexers and Demultiplexers -Memory and programming logic - HDL for combinational circuits.Synchronous Sequential Logic: Sequential circuits - Flip-flops - Analysis and design procedures - State reduction and state assignments - Shift registers - Counters - HDL for sequential logic circuits, shift registers and counters.Asynchronous Sequential Logic: Analysis and design of asynchronous sequential circuits - Reduction of state and flow tables - Race free state assignment - Hazards.

UNIT-IV

Microprocessor and Micro-controller - 8051 Micro-controller hardware: 8051oscillator and clock - Program counter and data pointer - A and B CPU register -Flags and PSW - Internal memory - Internal RAM - Stack and stack pointer - Special function registers - Internal ROM.Input / output pin, ports and circuits - External memory.Counter and Timer: Counter / Timer interrupts - Timing - Timer modes of operation - Counting.Serial data input / Output: Serial data interrupt - Data transmission - Data reception - serial data transmission modes.Interrupts: Timer flag interrupt - Serial port interrupt - External interrupt - reset -Interrupt control - Interrupt priority - Interrupt destination - Software generated interrupts.

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UNIT-V

Introduction - Addressing modes - Byte level logic operations - Bit level logic operations - Rotate and swap operations - Simple program.Arithmetic Operations: Introduction - Flags - Incrementing and Decrementing -Addition - Subtraction - Multiplication and Division - Simple Program.Introduction - External data move - code memory read only data move - PUSH and POP - Opcodes - Data exchange - Simple Programs.Jump and Call instructions: Introduction - Jump and call program range - Jumps -Calls and subroutine - Interrupt and returns - more detail on interrupts - Simple programs.Keyboard interfacing - Display interface - 7 segment and LCD display - D/A conversion - A/D conversion - Stepper motor Interface.

Text Books

1) Digital Logic and Computer Design - M. Morris Mano, Prentice Hall of India Private Limited.
2) Analysis and Modeling of Digital Systems - Zain Allabedin Navabee, 2/e , McGraw Hill Publishing Co. Ltd., New Delhi.
3) Digital Fundamentals - T.L. Floyd, 8/e, Pearson Education.
4) 1. The 8051 Microcontroller and Architecture, Programming and Applications -Kenneth J. Ayala, 2/e, Penram International.
5) The 8051 Microcontroller and Embedded System - Mohamed Ali maszidi & Janice Gillespie Maszidi, Pearson Education.
6) The 8051 Microcontroller and Architecture - Predko Mic, 2/e, Tata McGraw Hill Publishing Co. Ltd., New Delhi.
7) Electronic Instrumentation and Measurement Techniques - W.D. Cooper & A.D. Helfrick, Prentice Hall of India.
8) Electronic Instrumentation and Measurement - Kalasi.
9) A Course in Electrical and Electronic Measurement and Instrumentation - A.K. Sawhney, Dhanpat Rai and Sons.
10) Electronic Instrumentation and Measurements - P.B. Zbar, Mc Graw Hill International.
11) Measurement Systems Application and Design - Ernest O. Doebelin, 4/e, Tata McGraw Hill Publishing Co. Ltd.
12) Basic Electronics - A Text Lab Manual - Zbar, Malvino & Miller, Tata McGraw Hill Publishing Co. Ltd.
13) B.E.S. Practicals - R. Sugaraj Samuel & Horsley Solomon - Department of Electronic Science, C.T.M. College of Arts and Science, Chennai.

14) Fundamentals of Microprocessor 8085 - V. Vijayendran, S. Viswanathan

PRACTICAL-I year

1) adder, subtractor circuits and counters using logic gates.
2) pplication of microprocessor in basic mathematical function, code conversion and DAC.
3) Amplitude modulation and detection.
4) 2. Frequency modulation and detection.
5) 3. Pulse Amplitude modulation and detection.
6) Pulse Width modulation and detection.
7) Pulse Position modulation and detection.
8) Half, Full and BCD adders using simple logic gates.
9) Half, Full and BCD adders using NAND gates.
10) Half and Full subtractors using simple logic gates.
11) Half and Full subtractors using NAND gates.

12) Study of 7490 BCD counter, divided by (1 to 10) as scalar.
13) BCD to seven segment decoder using 7447/7448.
14) Microprocessor Practical Experiments
15) Addition, Subtraction, Multiplication and Division - 8 bit.
16) Picking up the largest/smallest in an array.
17) Ascending/Descending order.
18) Code conversions:
    a. Binary to BCD
    b. BCD to Binary
    c. Binary to ASCII
    d. ASCII to BinaryB.Sc. Electronics : Syllabus (CBCS)
    e. BCD to ASCII
    f. ASCII to BCD

Practicals II year

1) Construction of dual power supply using Zener diodes.
2) Construction of dual power supply using IC.
3) Op-amp - Inverting and Non-inverting modes, unity follower.
4) Op-amp - Summing amplifier - Inverting and Non-inverting modes.
5) Op-amp - Integrator and Differentiator.
6) Op-amp - Square wave generator.
7) Op-amp - Sine wave generator.
8) Instrumentation Amplifier.
9) Universality of NAND gate.
10) Universality of NOR gate.
11) Verification of basic Boolean identities using NAND gates.
12) Verification of basic Boolean identities using NOR gates.
13) Sum of Products and Product of Sums - NAND gates.
14) Sum of Products and Product of Sums - NOR gates.
15) Astable, Monostable multivibrators and Schmitt trigger using NAND gates.
16) Monostable multivibrators and Schmitt trigger using 555 timer.
17) Astable multivibrator using 555 timer.
18) Study of RS, D and JK flip flops.
19) AM, FM and PM modulation and detection techniques.
20) Characteristics of Zener diode.
21) Transistor characteristics in CE mode.
22) Regulated power supply using Zener diode.
23) Uses of CRO - Measurement of voltage, current, frequency and phase - Displaying
24) waveforms and Lissajou's figures - Study Experiment.
25) Transistor single stage amplifier - Frequency response.
26) Construction of low range power supply using rectifying diodes (6 V to 9 V).
27) Basic logic gates (AND, OR) using diodes.

28) Basic logic gates (AND, OR, NOT) using transistor.
29) characteristics of semiconductor devices such as UJT, JFET and SCR,
30) power control by SCR, audio wave generation and pulse shaping using Schmitt triggers.
31) Characteristics of UJT.
32) Characteristics of SCR.
33) SCR power control.
34) Characteristics of TRIAC.
35) Characteristics of JFET.
36) FET as an amplifier.
37) JFET multivibrator.
38) Emitter follower.
39) Darlington pair amplifier.
40) Transistor Hartley oscillator.
41) Transistor Colpitts oscillator.
42) Transistor phase shift oscillator.
43) Transistor Wien bridge audio oscillator.
44) Transistor monostable multivibrator.